

DMBI Module 3

Explain different types of attributes used in data exploration with examples.

1. Nominal Attributes:

- Categories with no order or ranking.
- Operations: equality, mode, contingency correlation.
- Examples: Gender (Male / Female), blood type (A / B / O / AB), country name.

2. Ordinal Attributes:

- Categories with a meaningful order but no fixed difference between values.
- Operations: equality, comparison, median, percentiles.
- Examples: Customer satisfaction (Poor / Average / Good), education level (High School < Bachelor < Master), shirt size (S / M / L).

3. Interval Attributes:

- Numeric with equal differences but no true zero point.
- Operations: addition, subtraction, mean, standard deviation.
- Examples: Temperature (°C or °F), calendar years (2020 , 2021 , 2022), IQ scores.

4. Ratio Attributes:

- Numeric with equal differences and a true absolute zero.
- Operations: all arithmetic (multiplication , division , ratios).
- Examples: Age (0 means none), height, weight, income, number of purchases.

What is an outlier? List types of outliers. Describe methods used for outlier analysis.

An outlier is a data point that differs significantly from other observations. It may indicate variability, measurement error, or novel behaviour.

Type	Description	Example
Point (Global) Outlier	Single value far outside normal range	Age = 150 in a dataset of human ages
Contextual (Conditional) Outlier	Abnormal only within a specific context	30°C in December (normal for July, outlier for winter)
Collective Outlier	A group of points deviating together	Three consecutive 0°C readings in a tropical climate

Methods:

1. Statistical (Z-Score) Method

- Assumes normal distribution. Calculates $Z = (x - \mu) / \sigma$.
- Rule: $|Z| > 3 \rightarrow$ outlier.
- Example: In test scores (mean=70, $\sigma=10$), a score of 30 has $Z = -4 \rightarrow$ outlier.

2. IQR (Interquartile Range) Method

- Non-parametric. Compute Q_1 (25th percentile), Q_3 (75th percentile).
- $IQR = Q_3 - Q_1$. Outlier if: $x < Q_1 - 1.5 \times IQR$ or $x > Q_3 + 1.5 \times IQR$.
- Example: $Q_1=50$, $Q_3=80$, $IQR=30$. Any value < 5 or > 125 is an outlier.

3. Visualization Methods

- Box Plot: Points beyond whiskers = outliers.
- Scatter Plot: Points far from main cluster.

4. Machine Learning Methods

- DBSCAN (Density-based): Points in low-density regions = outliers.
- Isolation Forest: Randomly isolates observations; outliers require fewer splits.

What is Noisy Data?

Random errors, distortions, or meaningless variations in data that obscure true patterns

- Causes: Faulty sensors, data entry errors, transmission glitches, measurement inaccuracies
- Difference from outliers: Noise is random error; outliers are valid but extreme values
- Examples: A temperature reading of 88.5°C in a room at 25°C; age entered as "abc"; static in an audio recording

Handling Noisy Data:

- Binning: Group values into bins; replace with bin mean, median, or boundaries
- Clustering: Group similar points; discard points not belonging to any cluster
- Regression: Fit a smoothing function; replace values with predicted points on curve
- Moving Average: Replace each value with average of its neighbouring values
- Manual Inspection: Expert review for critical datasets

Steps in data pre-processing

1. Data Cleaning

- Handle missing values (remove or impute).
- Remove duplicates and irrelevant data.
- Correct data entry errors.

2. Data Transformation

- Convert data types (e.g., strings to numbers, dates to datetime).
- Apply scaling/normalization (Min-Max, Z-score) for features with different units.
- Log or power transforms for skewed distributions.

3. Handling Outliers

- Detect outliers using IQR , Z-score , or visualization (boxplots , scatter plots).
- Remove, cap (winsorize), or treat outliers based on domain knowledge.

4. Encoding Categorical Variables

- Label encoding for ordinal categories (e.g., low / medium / high).
- One-hot encoding for nominal categories (e.g., red / green / blue → binary columns).
- Target encoding for high-cardinality features.

5. Feature Engineering & Selection

- Create new features from existing ones (e.g., age from birth date, ratios, interactions).
- Remove irrelevant or redundant features using correlation analysis, mutual information, or recursive elimination.

6. Data Splitting (for supervised learning)

- Split data into training, validation, and test sets (e.g., 70 / 15 / 15 or 80 / 20).
- Use stratified splitting for classification to preserve class balance.
- Ensure no data leakage (test data does not influence training).